Polynomial ModDiv Revisited

Exact Window Algebra, Rounded Residuals, and Separation from Middle-Product LWR

Samir Bouftass
Research article A3 v1.1 — November 2026

Abstract

Polynomial ModDiv mixes two distinct operations that must be kept separate: extraction of a coefficient window from an ordinary polynomial product, and coefficientwise modular quotient extraction. We first formalize an exact non-cyclic window model and prove a compatible-window identity for the common product. Over a finite field, the resulting unrounded exchange admits a polynomial-time passive predictor: any public solution of the linear window system suffices, even when it is not the planted secret. We then study the rounded publication appearing in the public multidimensional ModDiv proposal and retain its true deterministic remainders. This yields an exact residual identity, a circular-distance bound, and an explicit boundary criterion showing why a small residual does not by itself imply final-key equality. We also transcribe the local trailing-bit counter printed in the public proposal, including its preliminary integer division by n, and give exact counterexamples to the loop and prose readings that remain after that normalization. Finally, we separate this historical rounded construction from a published middle-product learning-with-rounding reference construction. For that reference we restate the exact rounding fibers, reconciliation condition, and a conditional classical-random-oracle IND-CPA theorem under its stated computational MP-CLWR source assumption. The comparison is deliberately negative in scope: algebraic similarity, commutativity, Toeplitz/Hankel structure, and the word "rounding" do not constitute a security reduction between the two distributions. No concrete security level or hardness theorem for historical polynomial ModDiv is claimed.

Keywords: modular quotient; polynomial windows; Toeplitz matrices; Hankel matrices; middle product; learning with rounding; MP-CLWR; key agreement; cryptanalysis.

Public record and scope

The public multidimensional ModDiv proposal is IACR ePrint 2021/701 [1]. It defines a non-cyclic polynomial window followed by coefficientwise integer reduction modulo an integer and integer division, and then applies a second window to cross-products. The public text also contains a local trailing-bit counter used in its final extraction. Its printed procedure first applies integer division by $2^{\log_2 n} = n$ to the intermediate integer and then runs the bit counter. We analyze those printed rules as mathematical objects; where the public text is asymmetric (it prints Alice's secret in Bob's later product), the symmetric reconstruction is stated explicitly and is not presented as proof of historical intent.

The comparison construction is the middle-product learning-with-rounding public-key encryption of Bai, Boudgoust, Das, Roux-Langlois, Wen and Zhang [2], with notation cross-checked against Boudgoust's public thesis [3]. The positive encryption theorem in this article concerns only that reference specialization under its stated computational MP-CLWR assumption and the classical random-oracle model. It does not transfer to the historical rounded exchange.

All mathematical claims needed below are proved in this article. No private report, chat, audit, thesis chapter, or unpublished project archive is used as a scientific reference.

1 Three objects that must not be conflated

The term *polynomial division* can refer to Euclidean division by a polynomial, whereas an integer division of each coefficient is a different operation. Their effects on algebra and distributions are different.

Label used in this article	Publication	Scope
P-WINDOW	An exact coefficient window of an ordinary product, over a specified field	An algebraic testbed, without coefficient rounding
P-ROUND	The same non-cyclic window, then canonical reduction modulo an integer and coefficientwise floor division	The publication public rounded-window form analyzed from [1]
P-REF	Nearest-rounded middle products with a long conditioned secret, plus a complete encryption mechanism	Published middle-product LWR reference construction, attributed to [2, 3]

These are organizational labels, not three newly claimed inventions. In particular, **P-WINDOW is not the historical publication with its integer rounding omitted from the explanation**: it is a deliberately different model used to show why window extraction alone cannot supply secrecy. P-ROUND does not become P-REF by renaming its factors.

We use fixed-length coefficient arrays in increasing degree order, with trailing zeros retained. The indeterminate is z. In the first part, n, m specify degrees and window endpoints, P, Q are positive integer arithmetic parameters, and p, q, when used for the historical conventions, are *bit lengths*. Section 11 introduces separate symbols p_0, q_0 for the two moduli of P-REF; they are not those bit lengths.

2 Monomial division and the exact public window

Let R be a commutative ring with identity. For integers $0 \le n < m$, define

$$C_{m,n}(F) = \sum_{i=n}^{m-1} [z^i]F \, z^{i-n}. \tag{1}$$

This equals the Euclidean quotient of F mod z^m by z^n. Both divisors are monic, so the definition is meaningful over such a ring without requiring inverses of arbitrary coefficients. There is no real-valued floor operation on polynomials in (1).

Write $X(z) = \sum_{j=0}^{n} x_j z^j$, allow a public polynomial A with coefficients a_i, and set

$$d = n + 1, \qquad L = m - n, \qquad M(A)_{ij} = a_{n+i-j}, \quad 0 \le i < L, \quad 0 \le j < d. \tag{2}$$

Proposition 1 — Window, Toeplitz and Hankel representations. The coefficient vector of $C_{m,n}(AX)$ is $M(A)x$. The matrix is Toeplitz. If J_d reverses a length-d vector and $H(A)_{ij} = a_{i+j}$, then

$$M(A) = H(A)J_d. \tag{3}$$

Only $a_0, \ldots, a_{m-1}$ are used. A coefficient a_m, even when nonzero and necessary to describe a polynomial of exact degree m, is irrelevant to this window.

Proof. The coefficient of degree $n+i$ in AX is $\sum_{j=0}^{n} a_{n+i-j}x_j$. Its smallest possible public index is zero and its largest is $m-1$. The entry depends on $i-j$. Reversing the columns of $H(A)$ gives $a_{i+d-1-j} = a_{n+i-j}$, proving (3). □

For example, when $m = 7, n = 2$, the output consists of the five coefficients of degrees 2 through 6, shifted down by two. The public matrix has size 5×3; its effective ribbon has seven coefficients, not fifteen independent entries.

The map $C_{m,n}$ is linear. It does not identify high degrees with low degrees as multiplication modulo $z^d - 1$ or $z^d + 1$ would. An irreducible polynomial modulus likewise defines a different algebra, not a harmless notation change.

3 Compatible nested windows preserve the common product

Assume $m > 2n$, and put $k = m - 2n$. For a vector v of length L, define

$$B_x(v)_i = \sum_{j=0}^{n} x_j v_{n+i-j}, \qquad 0 \le i < k. \tag{4}$$

The relevant indices satisfy $0 \le n + i - j \le L - 1$. In polynomial notation this is $C_{m,2n}(z^n X v)$.

Theorem 2 — Compatible-window identity. For $\deg X, \deg Y \le n$, over any commutative coefficient ring,

$$\begin{aligned} B_x(C_{m,n}(AY)) &= C_{m,2n}(AXY) \\ &= B_y(C_{m,n}(AX)). \end{aligned} \tag{5}$$

Proof. The i-th coefficient on the left is

$$\sum_{j=0}^{n} x_j [z^{2n+i-j}](AY).$$

For $0 \le i < k$, all indices $2n + i - j$ lie in $[n, m-1]$; these are exactly coefficients retained by the inner window. Hence the sum is $[z^{2n+i}](AXY)$. Commutativity exchanges X and Y, proving the other equality. □

This establishes the exact algebraic mechanism behind the compatible non-cyclic windows. It does not assert associativity for arbitrary windows. The shift z^n, the change from cutoff n to $2n$, and the output length k are part of the statement. The coefficient-index argument proves that every term needed by the final window survives the first extraction; it does not authorize other truncation patterns.

In the supplied algorithms all window dimensions are checked, including $n = 0$ in auxiliary algebra tests. The historical counter rules later in this article require $n \ge 1$.

4 Why the unrounded field exchange has a public predictor

In P-WINDOW, fix a known prime π and perform exact coefficient arithmetic in $\mathbb{F}_\pi$. The parties publish

$$U = C_{m,n}(AX), \qquad V = C_{m,n}(AY), \tag{6}$$

then compute the common vector in (5). No coefficientwise quotient or added noise occurs.

Theorem 3 — Efficient passive prediction for P-WINDOW. Given A, U, V from (6), an adversary can compute the common product vector in polynomial time in the input length. Full column rank is not required.

Proof. Solve $M(A)x^* = U$ over $\mathbb{F}_\pi$, choosing any solution by Gaussian elimination. A solution exists because the transcript is honestly generated. Its coefficient array defines a polynomial of degree at most n. By Theorem 2,

$$B_{x^*}(V) = B_y(C_{m,n}(Ax^*)) = B_y(U). \tag{7}$$

The right side is Bob's common value. Computing the left side uses only public data and x^*. Gaussian elimination and the polynomial operations use polynomially many field operations, each computable in time polynomial in $\log \pi$. The argument does not require recovering the original x, or finding a solution in its original restricted sampling support. □

A public deterministic derivation of that common vector is predictable as well. Extra secret-dependent state not specified in this model is not covered. The theorem does **not** permit solving the rounded equations by first pretending the missing remainders are zero.

Irreducibility alone does not rescue the corresponding *full, exact product*: over $\mathbb{F}_\pi[z]/(f)$ with irreducible f, a nonzero public multiplier is invertible, so an unrounded observation $U = AX \bmod f$ gives $X = A^{-1}U$. The polynomial Euclidean algorithm computes that inverse. This elementary statement is limited to the unrounded full product; it neither attacks a rounded window nor initiates the separate irreducible-polynomial research direction.

5 The rounded non-cyclic publication

Now return to integer coefficients. Fix $n \geq 1, m > 2n$, $P > Q \geq 1$, and use the dimensions in (2). Define

$$Z_X = M(A)x, \qquad U_i = F_{P,Q}(Z_{X,i}) = \left\lfloor \frac{[Z_{X,i}]_P}{Q} \right\rfloor, \tag{8}$$

with the analogous definition of V for y. This is P-ROUND. If $J = \lceil P/Q \rceil$, its output alphabet is $\{0, \ldots, J-1\}$. The last cell can be shorter than Q; no output-ring structure of size P/Q is assumed when this quotient is not integral.

The public proposal [1] requires $m > 2n+1$, so $L \geq d+1$ and $k \geq 2$, and specifies coefficient *bit lengths*. For the finite tests in this article, P, A_i have p bits and Q, X_j, Y_j have q bits. Thus

$$\begin{aligned} 2^{p-1} &\leq P, A_i < 2^p, \\ 2^{q-1} &\leq Q, X_j, Y_j < 2^q. \end{aligned} \tag{9}$$

Consequently, A_i can exceed P, and X_j can exceed Q. "Pseudorandom" in the source is not a complete probability law; uniform independent draws in these finite intervals are a declared testing convention, not a retrospective specification of an unknown generator.

Proposition 4 — Coefficient-size bound. Let p, q be positive integers. If $0 \le a_i < 2^p$ and $0 \le x_j < 2^q$, every retained ordinary product coefficient is at most

$$d(2^p - 1)(2^q - 1) < d\,2^{p+q}. \tag{10}$$

It has at most $p + q + \lceil \log_2 d \rceil$ bits, with the bit length of zero taken to be zero. More generally, a sum of d nonnegative s-bit-bounded integers needs at most $s + \lceil \log_2 d \rceil$ bits.

Proof. Bound each nonnegative summand, sum the d bounds, and use $d \le 2^{\lceil \log_2 d \rceil}$. Strict inequality with the next power of two gives the stated bit length. □

Here $d = n + 1$, not n. This is why replacing the bit length of a sum by a product such as $s \log_2 n$ is not a valid general rule. No carry propagates between formal polynomial coefficients, but ordinary integer carries and the bound (10) still occur *within* each coefficient.

6 What cancellation leaves after coefficient rounding

Define true Euclidean remainders by

$$Z_X = Pz_X + QU + e_X, \qquad 0 \le e_{X,i} < \min(Q, P - QU_i), \tag{11}$$

and similarly for Y. These are deterministic functions of the original product; no independence assumption is introduced.

Set

$$R_A = B_x(V), \quad R_B = B_y(U), \qquad S_A = [QR_A]_P, \quad S_B = [QR_B]_P. \tag{12}$$

All modular operations on vectors are componentwise.

Theorem 5 — Rounded residual identity and circular bound. For every $0 \le i < k$,

$$S_{A,i} - S_{B,i} \equiv \varepsilon_i := \sum_{j=0}^{n} (y_j e_{X,n+i-j} - x_j e_{Y,n+i-j}) \pmod{P}. \tag{13}$$

If the secret coefficients are nonnegative, then

$$d_P(S_{A,i}, S_{B,i}) \le |\varepsilon_i| \le (Q - 1)\max(\|x\|_1, \|y\|_1). \tag{14}$$

In particular, $x_j, y_j \le b$ implies the upper bound $db(Q - 1)$.

Proof. Substitute (11) into $QB_x(V)$ and $QB_y(U)$. The principal terms cancel by Theorem 2; the terms multiplied by P disappear modulo P, leaving (13) with the displayed sign. Each of the nonnegative sums $B_y(e_X)_i$ and $B_x(e_Y)_i$ is bounded by $(Q - 1)\|y\|_1$ and $(Q - 1)\|x\|_1$, respectively. The absolute difference is bounded by their maximum. Circular distance is at most the magnitude of any congruent integer displacement. □

This statement allows $Q \nmid P$. In that case $e_{X,i}$ cannot generally be replaced by $[Z_{X,i}]_Q$. It bounds circular distance between residues, not an ordinary difference of canonical representatives and not equality after a final quotient. The general principles developed in earlier scalar and distributional analyses apply, but their scalar probability constants are not automatically vector probability constants.

7 A public fixed divisor still needs a boundary argument

A later design may fix a public divisor D before the exchange. This is distinct from selecting separate local divisors. It is useful to isolate what such a choice proves. Let

$$K_{A,i} = \lfloor S_{A,i}/D \rfloor, \qquad K_{B,i} = \lfloor S_{B,i}/D \rfloor. \tag{15}$$

Proposition 6 — Exact common-divisor criterion when $D \mid P$. Write $P = DJ$, $S_{B,i} = Dv_i + \rho_i$, $0 \le \rho_i < D$, and use the displacement ε_i of (13). Then

$$K_{A,i} - K_{B,i} \equiv \left\lfloor \frac{\rho_i + \varepsilon_i}{D} \right\rfloor \pmod{J}. \tag{16}$$

If $J \ge 2$ and $|\varepsilon_i| < D$, equality holds exactly when $0 \le \rho_i + \varepsilon_i < D$.

Proof. Divide $\rho_i + \varepsilon_i$ by D, then reduce its quotient modulo J; this gives (16). Under the additional displacement bound its quotient belongs to $\{-1, 0, 1\}$. Only zero preserves a quotient modulo $J \ge 2$. □

Without the divisibility assumption, the last quotient cell may be incomplete; equation (16) is not used. Small displacement alone does not suffice in that case either. A selected example in the original bit-size domain has

$$\begin{gathered} n = 2,\ m = 6,\ p = 11,\ q = 3,\ P = 1543,\ Q = 5,\ D = 256, \\ A = (1373, 1276, 1798, 1379, 1431, 1938, 1351), \\ x = (4, 6, 6), \qquad y = (7, 6, 7). \end{gathered} \tag{17}$$

Exact computation gives

$$\begin{aligned} U &= (296, 163, 19, 293), & V &= (107, 11, 3, 189), \\ R_A &= (720, 840), & R_B &= (3183, 3306), \\ S_A &= (514, 1114), & S_B &= (485, 1100), \\ K_A &= (2, 4), & K_B &= (1, 4). \end{aligned} \tag{18}$$

The first-coordinate circular separation is 29, much less than 256, but the boundary at 512 is crossed. This is a finite diagnostic of this common divisor, not a disproof of every fixed-public-r construction or of a separately proved probabilistic bound. We do not change the public, preselected r.

8 Historical local-counter readings: normalization, defects, and erasure

For documentary clarity, this section analyzes two exact symmetric readings of the local trailing-bit counter printed in the public proposal [1]. They are historical rules, not a recommended reconciliation mechanism. The public proposal prints X in Bob's further computation, although Bob has Y; H0/H1 therefore make the symmetric reconstruction explicit by using Bob's own secret.

The printed procedure also performs an integer division before the bit loop. Since $2^{\log_2 n} = n$, define

$$\widetilde{R}_i = \left\lfloor \frac{R_i}{n} \right\rfloor.$$

For a positive integer w, write $\tau(w)$ for the length of its final run of identical binary digits. The printed scalar loop returns $\tau(w)-1$, whereas the prose suggests $\tau(w)$. On zero the unbounded-integer loop does not terminate. Hence the historical counter is defined only when every required $\widetilde{R}_i > 0$.

The two candidate outputs are

$$\begin{aligned} \widetilde{K}_i^{(0)} &= \left\lfloor \frac{[QR_i]_P}{n2^{2q+\tau(\widetilde{R}_i)}} \right\rfloor, \\ \widetilde{K}_i^{(1)} &= \left\lfloor \frac{[QR_i]_P}{n2^{2q+\tau(\widetilde{R}_i)+1}} \right\rfloor. \end{aligned} \tag{19}$$

Here H0 is the loop reading and H1 is the prose reading. The normalization affects only the counter input: the numerator remains the canonical residue $[QR_i]_P$.

Counterexample 7 — Non-cyclic normalized-counter disagreement. With the dimensions and moduli in (17), take

$$\begin{aligned} A &= (1580, 1057, 1283, 1933, 1248, 1713, 1749), \\ x &= (4,4,5), \qquad y = (5,4,4). \end{aligned} \tag{20}$$

The two historical readings both disagree, although P is prime, the public matrix has full column rank, and its effective ribbon is not d-periodic modulo P.

Exact verification. The publications are $U = (57, 235, 124, 290)$ and $V = (306, 101, 117, 246)$. The intermediate values are

$$\begin{aligned} R_A &= (2402, 1957), \quad R_B = (1788, 2886), \\ S_A &= (1209, 527), \quad\ S_B = (1225, 543). \end{aligned}$$

The normalized counter inputs are $(1201, 978)$ and $(894, 1443)$, with terminal-run lengths $(1,1)$ and $(1,2)$. H0 gives $(4,2)$ and $(4,1)$; H1 gives $(2,1)$ and $(2,0)$. The leading 3×3 minor has determinant 1211 modulo 1543. Trial division through the primes at most 37 verifies primality. Direct convolution verifies all outputs. □

A second localized witness shows that the normalized local-counter output is not a function of the final residue alone. Keep the same public ribbon as in (20), but take

$$x = (4,4,4), \qquad y = (4,5,7). \tag{21}$$

Exact computation gives

$$\begin{aligned} U &= (50, 23, 176, 212), \quad V = (283, 297, 98, 78), \\ R_A &= (2712, 1892), \quad R_B = (1169, 1889), \\ S_A &= (1216, 202), \quad S_B = (1216, 187). \end{aligned}$$

At the first coordinate the final residues are equal, but

$$\tau(2712 \operatorname{div} 2) = \tau(1356) = 2, \qquad \tau(1169 \operatorname{div} 2) = \tau(584) = 3.$$

Consequently H0 gives $(2,0)$ and $(1,0)$, while H1 gives $(1,0)$ and $(0,0)$. Thus equal final residues do not force equal local-counter outputs under the printed normalized rule.

Proposition 8 — Every defined output is erased at the minimal precision. For H0, if $P \le n2^{2q+1}$, every *defined* output in (19) is zero. For H1 it suffices that $P \le n2^{2q+2}$. If p is the least integer strictly exceeding $2q + \log_2 n$ and P has p bits, both erasure conclusions apply to all defined outputs.

Proof. A defined counter requires $\widetilde{R}_i = \lfloor R_i/n \rfloor \ge 1$, hence $\tau(\widetilde{R}_i) \ge 1$, while $[QR_i]_P < P$. Each stated bound therefore places the numerator strictly below the denominator. The minimal bit length is $p = 2q + \lfloor \log_2 n \rfloor + 1$, whence $P < 2^p \le n2^{2q+1}$. Inputs with $0 \le R_i < n$ give $\widetilde{R}_i = 0$, for which the printed unbounded loop is undefined; such a non-computation is never counted as agreement. □

The zero-input issue occurs inside the historical bit-size domain. For example, with

$$n = 2,\ m = 6,\ p = 4,\ q = 1,\ P = 11,\ Q = 1,$$

$$A = (10, 11, 13, 9, 11, 14, 8), \qquad x = y = (1, 1, 1),$$

one obtains $U = V = (1, 0, 0, 1)$ and $R_A = R_B = (1, 1)$. Thus $\lfloor R_i/n \rfloor = 0$, so the normalized historical counter is undefined even though $R_i > 0$.

For conditional finite checks, we retain two selected public ribbons. The first is the ribbon in (20); the second is

$$A^{(2)} = (1810, 1110, 2023, 1815, 1926, 1554, 2031).$$

Enumerating all $64^2 = 4096$ secret pairs $x, y \in \{4, 5, 6, 7\}^3$ gives H0 agreement in 944 pairs and H1 agreement in 1600 pairs for the first ribbon, and H0 agreement in 986 pairs and H1 agreement in 1606 pairs for the second. No normalized counter is undefined in these two enumerations. These are conditional counts at deliberately selected public values, not averages over the entire public distribution, and no extrapolation to the full public-parameter law is made.

9 The rounded inverse problem and boundary projections

For a public vector u with every coordinate in $\{0, \dots, \lceil P/Q \rceil - 1\}$ (reject other inputs), known integer bounds $\ell \le x_j \le b$, and a fixed public matrix, the rounded inverse relation is exactly the existence of integer z_i such that

$$Qu_i \le \sum_j M(A)_{ij} x_j - Pz_i \le \min\{Q(u_i + 1) - 1, P - 1\}, \qquad \ell \le x_j \le b. \tag{22}$$

Indeed the middle expression is a canonical residue in precisely the requested cell. Conversely, any compatible secret supplies the integer quotients z_i. This formulation distinguishes finding an admissible preimage from recovering the original sampled secret. Writing it as integer feasibility does not give a polynomial-time algorithm when the dimensions grow. The scalar interval algorithm from the scalar setting is not invoked as a vector solver.

Non-cyclic structure also does not exclude every projection relation.

Proposition 9 — Exact all-ones projection criterion. Define column sums $s_j = \sum_{i=0}^{L-1} M(A)_{ij}$. Then

$$s_j - s_{j+1} = a_{m-1-j} - a_{n-1-j}, \qquad 0 \le j < n. \tag{23}$$

All column sums coincide modulo P exactly when the n displayed boundary differences vanish modulo P. If the effective public coefficients are independent uniforms in an interval of $H \le P$

consecutive integers, and $m > 2n$, the probability of these equalities is exactly H^{-n}.

Proof. In the two column sums all common interior coefficients cancel, leaving their two endpoints. Under the dimension assumption the lower and upper endpoint index sets are disjoint, and all indices within each set are distinct. Reduction modulo P is injective on an interval of length at most P. Each pair therefore matches with probability $1/H$, independently of the other pairs. □

This is a result about one predetermined projection weight vector, not about every relation chosen after observing A, and not a claim of either general insecurity or security. No LLL, BKZ, Babai, or mixed-integer benchmark is claimed in this article; the proposition concerns only the displayed predetermined projection.

10 Why algebraic similarity does not identify the source distribution

Proposition 10 — Two exact statistical distances. Compare the following auxiliary public distributions.

First, sample a uniformly random Toeplitz $L \times d$ matrix over $\mathbb{Z}/P\mathbb{Z}$ through its $m = L + d - 1$ independent diagonal coefficients. Its distance from a matrix of Ld independent uniform entries is

$$1 - P^{m-Ld}. \tag{24}$$

Second, let P have p bits, put $H = 2^{p-1}$, and sample an effective ribbon of length m uniformly from $[H, 2H)^m \cap \mathbb{Z}^m$. After reduction modulo P, its distance from an independent full-modulus ribbon is

$$1 - (H/P)^m. \tag{25}$$

Proof. In the first comparison, the map from the diagonal ribbon to its matrix is injective, because every diagonal position occurs. The structured law is uniform on P^m of the P^{Ld} matrices. In the second, the interval has $H \le P$ distinct residues, so the law is uniform on H^m of P^m ribbons. For a uniform law on a subset of size s of an ambient set of size N, direct summation of total variation gives $1 - s/N$. □

When $L = 1$ or $d = 1$, (24) is zero, as it should be. Neither expression is a cryptanalytic advantage against a correctly defined rounded-publication game: the public distribution must be the same in that game's real and ideal worlds. The distances instead prohibit a silent statistical replacement of the public matrix.

A similar issue affects the factors. In P-ROUND, the public factor is long and the private factor short; the published MP-CLWR source used below has a short public factor and a long conditioned secret. In the direct historical identification, assigning $N = d$ and middle-product length $D = L$ would even violate $D \le N$ under $L \ge d + 1$. Commutativity does not change what an adversary knows. This rules out that simple identification, not every possible reduction. No general hardness bridge is claimed here.

11 Reference rounding, inverse sampling and reconciliation

We now describe **P-REF**, an even-modulus specialization of the published middle-product LWR construction [2, 3]. All symbols in this section are separate from the historical bit lengths. Let

$$q_0 = 2^b, \quad p_0 = 2^a, \quad b > a \ge 1, \quad b \ge 3, \qquad B = q_0/p_0. \tag{26}$$

In particular B is even and $8 \mid q_0$. Define nearest rounding with exact half-integers rounded upward:

$$\mathcal{R}(c) = \left\lfloor \frac{p_0[c]_{q_0}}{q_0} + \frac{1}{2} \right\rfloor_{p_0}. \tag{27}$$

Proposition 11 — Shifted quotient and exact inverse fiber. For all integers c,

$$\mathcal{R}(c) = F_{q_0,B}(c + B/2). \tag{28}$$

For $y \in \{0, \dots, p_0 - 1\}$, a uniform inverse-fiber sample is

$$\mathrm{INV}(y) = [By - B/2 + e]_{q_0}, \qquad e \leftarrow \mathrm{Unif}\{0, \dots, B-1\}. \tag{29}$$

Any two points in one fiber have circular distance at most $B - 1$.

Proof. Write $[c]_{q_0} = Bu + r$, $0 \le r < B$. Both rounding expressions give $[u + \mathbf{1}_{r \ge B/2}]_{p_0}$. Formula (29) lists exactly the B consecutive circular representatives belonging to a fiber, each once. Two lifts in that interval differ by at most $B - 1$. □

Nearest rounding is not the unshifted floor in P-ROUND: $q_0 = 16, p_0 = 4, c = 2$ gives 1 here and 0 for the unshifted quotient. Boundary ties have positive probability. Replacing INV by By also changes the law.

For a canonical $x \in \mathbb{Z}_{q_0}$, define

$$K(x) = \left\lfloor \left\lfloor \frac{2x}{q_0} + \frac{1}{2} \right\rfloor \right\rfloor_2, \qquad C(x) = \left\lfloor \left\lfloor \frac{4x}{q_0} \right\rfloor \right\rfloor_2. \tag{30}$$

Put

$$\begin{aligned} I_0 &= \{0, \dots, q_0/4 - 1\}, \\ I_1 &= \{-q_0/4, \dots, -1\}, \\ E &= \{-q_0/8, \dots, q_0/8 - 1\}. \end{aligned}$$

Define $\mathrm{REC}(y, c) = 0$ if $[y]_{q_0} \in I_c + E \pmod{q_0}$, and 1 otherwise. All intervals contain integers; the endpoints are part of the specification.

Lemma 12 — Reconciliation and uniformity. If $d_{q_0}(x, y) < q_0/8$, then

$$\mathrm{REC}(y, C(x)) = K(x). \tag{31}$$

For uniform x, the pair $(K(x), C(x))$ is uniform on four pairs.

Proof. The four successive quarters have pairs $(0,0), (1,1), (1,0), (0,1)$, each on exactly $q_0/4$ integers. For correction, write $q_0 = 8h$. For hint 0 the zero-decoding region is $[-h, 3h - 2] \pmod{8h}$, and for hint 1 it is $[-3h, h - 2] \pmod{8h}$. The quarters with key bit zero, expanded by integer errors in $[-h + 1, h - 1]$, stay in their respective zero regions; the other two expanded quarters stay outside them. This also checks the modular endpoint and $h = 1$. □

These operations are attributed to the reference construction and its reconciliation lineage. They are not added silently to the historical exchange.

12 Complete P-REF algorithms and perfect correctness

Use integers $d_0, k_0 \geq 2$, $N \geq d_0 + k_0$, $t \geq 1$, and put $D_0 = d_0 + k_0$. For fixed array lengths l_a, l_b, define

$$\mathrm{MP}_D(a,b)_i = [z^{h+i}](ab), \qquad h = (l_a + l_b - 1 - D)/2, \tag{32}$$

only when h is a nonnegative integer. Products are ordinary convolutions, not cyclic products. The random oracle is classical and has domain $\{0,1\}^{d_0}$ and range $\{0,1\}^{k_0}$.

KeyGen. Sample s uniformly from $\mathbb{Z}_{q_0}^{N+D_0-1}$, rejecting until the $D_0 \times N$ Hankel matrix (s_{i+j}) is surjective. For each $j = 1, \ldots, t$, draw an independent uniform $a_j \in \mathbb{Z}_{q_0}^N$, set

$$b_j = \mathcal{R}(\mathrm{MP}_{D_0}(a_j, s)), \tag{33}$$

and return $pk = ((a_j, b_j))_j$, $sk = s$.

Over $\mathbb{Z}/2^b\mathbb{Z}$, surjectivity is equivalent to full row rank modulo 2. One direction follows by reduction. Conversely a full-rank binary minor has odd determinant, hence is invertible modulo 2^b, giving a right inverse on those columns. A nonzero determinant over the reals is insufficient; the matrix [2] modulo 4 is the simplest counterexample. The published random-Hankel bound in [3, Lemma 4.1] gives acceptance probability at least $1/2$ here. Thus exact rejection samples the required conditional law in expected polynomial time; a cap of λ trials causes an explicit abort of probability at most $2^{-\lambda}$. The density bound is an attributed published input, not proved anew by the finite tests.

Encrypt. For each j, draw independent uniform $r_j \in \{0,1\}^{k_0+1}$ and independently lift each coefficient of b_j by (29), obtaining u_j. Compute

$$c_1 = \sum_j r_j a_j \pmod{q_0}, \qquad v = \sum_j \mathrm{MP}_{d_0}(r_j, u_j) \pmod{q_0}, \tag{34}$$

then, for a k_0-bit message m, return

$$(c_1, c_2, c_3) = (c_1, C(v), \mathcal{H}(K(v)) \oplus m). \tag{35}$$

Here c_1 has length $N + k_0$, c_2 length d_0, and c_3 length k_0. The maps $K, C, \mathcal{R}, \mathrm{INV}$ act coefficientwise.

Decrypt. Compute

$$w = \mathrm{MP}_{d_0}(c_1, s) \pmod{q_0}, \qquad m' = c_3 \oplus \mathcal{H}(\mathrm{REC}(w, c_2)). \tag{36}$$

The scheme is randomized even though its public-key rounding is deterministic. A real hash function is not silently identified with $\mathcal{H}$.

Lemma 13 — Reference weak associativity. At these exact lengths,

$$\mathrm{MP}_{d_0}(r, \mathrm{MP}_{D_0}(a, s)) = \mathrm{MP}_{d_0}(ra, s). \tag{37}$$

Proof. The inner product window starts at degree $N-1$; the outer window starts at degree k_0. Its i-th coefficient is $\sum_{h=0}^{k_0} r_h [as]_{N+k_0-1+i-h}$. These indices all lie between $N-1$ and $N+D_0-2$, so no needed coefficient was discarded. They are exactly the coefficients selected from $(ra)s$. □

Theorem 14 — Perfect correctness of P-REF. If

$$p_0 > 8t(k_0 + 1), \tag{38}$$

then decryption returns every message for every honest key and every encryption randomness.

Proof. Let e_j be a signed circular lift of $u_j - \mathrm{MP}_{D_0}(a_j, s)$ with every coefficient bounded in magnitude by $B - 1$. Such lifts exist by Proposition 11 because both vectors round to b_j. By (37),

$$v - w \equiv \sum_j \mathrm{MP}_{d_0}(r_j, e_j) \pmod{q_0}. \tag{39}$$

Each output is a sum of at most $t(k_0 + 1)$ signed error terms, with binary multipliers. It has a representative of magnitude at most $t(k_0 + 1)(B - 1) < q_0/8$ under (38). Lemma 12 recovers $K(v)$ exactly. The two oracle masks coincide and cancel. □

This is a theorem about the specified reference, not a reconciliation theorem for H0/H1 or an automatic repair of P-ROUND.

13 The precise computational assumption

The source assumption for P-REF is $\mathrm{MP^{\not|}CLWR}_{p_0,q_0,N,D_0,t}$ in its computational, task-preservation formulation [2, 3]. The real source samples a single s from the conditioned law in KeyGen and then the t pairs in (33). The ideal source samples each a_j uniformly as before and sets $b_j = \mathcal{R}(y_j)$ for independent uniform $y_j \in \mathbb{Z}_{q_0}^{D_0}$. In this divisible-modulus specialization, b_j is uniform in $\mathbb{Z}_{p_0}^{D_0}$.

For each polynomial-time challenger that maps its source and independent coins to an input and a target, the assumption says: if every polynomial-time adversary has negligible target-recovery probability under the ideal source, then every such adversary has negligible recovery probability under the real source. The challenger and adversary algorithms must be the same in both worlds. The source secret is not supplied to that challenger as extra information. The sample budget t and the full parameter family are fixed in advance.

This is not asserted to be ordinary decisional pseudorandomness of the public key. Nor is it an assumption that the encryption scheme is already secure. Section 14 exhibits the specific recovery task and proves its ideal-world bound. Public statements on MP-LWE hardness in [2] concern additional theorems with their own distributions and parameters; that harder source-justification chain is not reconstructed or extended to historical MODDIV in this article.

14 Conditional classical random-oracle security of the reference

This section gives a self-contained proof for the stated specialization while retaining attribution to the published construction. All dimensions and modulus bit lengths below are polynomial in a security parameter λ, and all adversaries make polynomially many classical oracle queries.

Lemma 15 — Universal hashing with the complete lift as seed. Let the u_j be independent uniform vectors in $\mathbb{Z}_{q_0}^{D_0}$. For binary arrays $r = (r_1, \ldots, r_t)$, define

$$h_u(r) = \sum_j \mathrm{MP}_{d_0}(r_j, u_j) \pmod{q_0}. \tag{40}$$

For every $r \neq r'$, the difference $h_u(r) - h_u(r')$ is uniform in $\mathbb{Z}_{q_0}^{d_0}$. In particular its zero probability is $q_0^{-d_0}$.

Proof. In one nonzero difference block $\delta = r_j - r'_j$, let h be its last nonzero coefficient. It is

1 or -1. In the i-th middle-product output it multiplies u_{k_0+i-h}. Select these d_0 coefficients as variables and fix all other coefficients and blocks. Solving outputs from the largest i to the smallest is triangular: earlier coefficients of δ involve only larger selected indices or already fixed coefficients. The diagonal is δ_h, a unit even modulo a power of two. Each output has the same number of preimages, proving uniformity. □

Lemma 16 — Ideal lifted source with bounded leakage. In the ideal source, generate each u_j first uniformly in $\mathbb{Z}_{q_0}^{D_0}$ and set $b_j = \mathcal{R}(u_j)$. This gives exactly the joint distribution obtained by sampling b_j and then applying INV. With independent uniform binary r_j, define c_1, v by (34). If V_0 is an independent uniform vector of length d_0, then

$$\mathrm{TV}\big((a,u,c_1,v),(a,u,c_1,V_0)\big) \le \epsilon_{\mathrm{ext}} := \frac{1}{2}\sqrt{2^{-t(k_0+1)} q_0^{N+k_0+d_0}}. \tag{41}$$

Proof. Proposition 11 shows that all fibers have size B; sampling their label uniformly and then a uniform element of the fiber is equivalent to sampling the full element first. Thus the complete seed u is uniform and independent of a, r.

Fix a. The map $r \mapsto c_1$ has at most $q_0^{N+k_0}$ values. Conditional on a nonempty fiber of size s_c, the binary array r is uniform on that fiber. Universality implies expected squared distance of its hashed distribution from uniform is at most $1/s_c$: expand the collision probability over equal and unequal input pairs, using collision probability $q_0^{-d_0}$ for the latter. Cauchy–Schwarz converts this into an average total variation bound $\frac{1}{2}\sqrt{q_0^{d_0}/s_c}$, with the seed included in the observation. If $T = 2^{t(k_0+1)}$, weighting by s_c/T and applying Cauchy–Schwarz once more gives

$$\frac{\sqrt{q_0^{d_0}}}{2T}\sum_c \sqrt{s_c} \le \frac{1}{2}\sqrt{\frac{q_0^{d_0} q_0^{N+k_0}}{T}}.$$

This holds for every fixed a, so averaging proves (41). □

Revealing the whole lift is stronger than revealing only b. Apply $b = \mathcal{R}(u)$, discard u, and compute $(C(v), K(v))$; data processing does not increase the distance. In the comparison world, Lemma 12 makes $K(V_0)$ uniform even given $C(V_0)$. Hence every recovery strategy for $z = K(v)$ from (pk, c_1, c_2) has ideal-source success at most

$$2^{-d_0} + \epsilon_{\mathrm{ext}}. \tag{42}$$

Theorem 17 — Conditional IND-CPA in the classical ROM. Suppose

$$\begin{gathered} d_0 \ge \lambda, \qquad t(k_0+1) \ge 2\lambda + (N + k_0 + d_0)\log_2 q_0, \\ p_0 > 8t(k_0+1). \end{gathered} \tag{43}$$

Under the computational source assumption of Section 13, P-REF is IND-CPA secure in the classical random-oracle model. Exact KeyGen has expected polynomial running time; capped KeyGen has an explicit negligible abort allowance.

Proof. The ideal recovery bound (42) is at most $2^{-\lambda} + 2^{-\lambda-1}$, hence negligible. The recovery challenger takes the source pairs, draws the same r_j and INV samples as Encrypt, outputs (pk, c_1, c_2), and keeps target $z = K(v)$. It is a valid challenger for the computational assumption: it uses no source secret and its procedure is identical in both worlds. The assumption therefore makes every polynomial-time real-source recovery probability negligible.

It remains to link IND-CPA advantage to this recovery task. Given (pk, c_1, c_2), an extractor

simulates an IND-CPA adversary, a lazy random-oracle table, and an independent challenge bit. The adversary sees pk, chooses its two messages, and receives (c_1, c_2, c_3) with c_3 uniform and independent. The challenger may presample c_1, c_2, z, because their law does not depend on the message. In this independent-mask game the transcript does not reveal the challenge bit.

Couple it to the real game until the first query on z. Before that event, the real value $\mathcal{H}(z)$ is an unused uniform mask and all observations have the same law. Thus the adversary's success bias $\eta = |\Pr[\widehat{\beta} = \beta] - 1/2|$ is at most the probability of querying z. Record all distinct queries, including those made before the challenge. If their total is bounded by a public polynomial $T_{\mathcal{H}} \geq 1$, output a uniformly selected position among $T_{\mathcal{H}}$ slots, using a fixed dummy candidate for an unused slot. Recovery then succeeds with probability at least $\eta/T_{\mathcal{H}}$. The extractor does not recognize the correct query and does not identify it by a possibly colliding oracle output.

A nonnegligible IND-CPA bias would contradict real-source target hardness. This proves the claim for classical queries. The conditioned secret sampler uses polynomial-time rank tests; the published acceptance bound described in Section 12 gives expected polynomial time, or a negligible difference under the stated cap. Correctness follows separately from Theorem 14. □

The explicit condition $d_0 \geq \lambda$ is sufficient to make guessing the oracle input negligible. The proof would not justify dropping that requirement while leaving d_0 constant. This is a conditional theorem for an attributed reference scheme. It is not a proof for a concrete hash, the quantum random-oracle model, chosen-ciphertext security, or historical P-ROUND.

15 Nonempty parameter families and representation costs

The conditions are not merely a formal conjunction with no examples. The following conservative family is included only to demonstrate simultaneous satisfiability of the displayed inequalities; it is not optimized for deployment.

Proposition 18 — A polynomially describable family satisfying (43). For integer $\lambda \geq 2$, let $L_0 = \lfloor \log_2 \lambda \rfloor + 1$, set

$$N = 2\lambda, \quad d_0 = k_0 = \lambda, \quad t = 64L_0, \tag{44}$$

and choose p_0 as the least power of two strictly greater than $8t(\lambda + 1)$, and q_0 as the least power of two strictly greater than $2p_0N^3t$. Then all the dimension, modulus, correctness and entropy inequalities above hold. This statement does not establish the source hardness assumption.

Proof. Strict powers give $p_0 > 8t(\lambda + 1)$, $q_0 > p_0$, and $8 \mid q_0$. Also $D_0 = N$. The upper bounds

$$p_0 < 32t\lambda, \qquad q_0 < 1024t^2\lambda^4 = 2^{22}L_0^2\lambda^4 \tag{45}$$

imply $\log_2 q_0 < 22 + 6L_0$. For $L_0 \geq 3$,

$$2 + 4\log_2 q_0 < 90 + 24L_0 \leq 64L_0 = t.$$

Multiplying by λ proves the entropy inequality in (43), since $t(\lambda + 1) \geq t\lambda$. For $L_0 = 2$, $\lambda = 2, 3$; direct integer calculation gives respectively $(t, p_0, q_0) = (128, 4096, 134217728)$ and $(128, 8192, 536870912)$, both satisfying it. All dimensions, moduli and bit lengths have polynomial upper bounds. □

Writing $\lambda = 128$ in this family is not an estimate of 128-bit resistance. Source hardness, reduction losses, best attacks and concrete parameters would still need a separate study. Here the extra size of q_0 preserves the conservative family; it is not used to claim a new hardness reduction.

With fixed-length, bit-packed fields, P-REF has arithmetic payload sizes

$$\begin{aligned} |pk| &= t(N \log_2 q_0 + D_0 \log_2 p_0), \\ |sk| &= (N + D_0 - 1) \log_2 q_0, \\ |ct| &= (N + k_0) \log_2 q_0 + d_0 + k_0 \end{aligned} \tag{46}$$

in bits, excluding profile identifiers and formatting. These describe the explicitly listed public polynomials, not a seed-compressed variant. For historical P-ROUND, effective public data include m ribbon coefficients and L quotients; their sizes follow their own alphabets, not (46). An unused a_m may be omitted only from an explicitly documented representation, not from a claim about the original file format.

16 Reproducible checks and limits of the implementation

The package uses standard-library Python, arbitrary-precision integers, exact fractions where probabilities are computed, and deterministic test seeds. The mathematical reference exposes separate functions for windows, historical rounding, the bare finite-field solver and P-REF. The seeded generator and the lazy ideal-oracle table are testing facilities, not a deployable cryptographic implementation or a proof about a physical entropy source.

Verification family	Extent executed
Signed window, matrix and nested-product instances	2,500
Exact finite-field key predictions	10,913
Predictions among those with rank-deficient public matrices	1,817
Rounded residual and circular-bound instances	4,000
Column-sum boundary identities	4,000
Secret pairs at the two selected fixed public ribbons	8,192
Defined historical erasure checks	7,425
Normalized-zero undefined-counter checks	50
Coefficient bit-length bounds	361
Signed nearest-rounding inputs	12,315
Complete inverse-rounding fibers	492
Reconciliation inputs with all permitted small displacements	21,336
Binary rank versus exhaustive images modulo 4	64
Reference weak-associativity instances	1,200
Nonzero binary-difference hash cases modulo 4	26
Complete seeds examined in those hash cases	6,656
Reference encryptions	132
Additional extremal-lift encryptions	2

Verification family	Extent executed
Parameter-family inequalities for $2 \le \lambda \le 256$	255

The categories are not summed as independent cryptographic experiments. The 1,817 rank-deficient predictions are a subset of the 10,913, not additional trials. The statement about primes in the finite-field solver is an input assumption; that solver is not used over arbitrary composite moduli.

The 132 reference encryptions consist of 120 functionality tests at $(N, d_0, k_0, t, p_0, q_0) = (8, 4, 4, 3, 256, 65536)$, which satisfies the correctness condition but not the security entropy bound, and four tests each for the conservative family at $\lambda = 2, 8, 16$. All messages are recovered. These finite labels do not confer security. The largest observed error at the functionality profile is 797, below the proved bound $3 \cdot 5 \cdot 255 = 3825 < 8192$; a bound is justified by Theorem 14, not by this maximum.

A negative control uses $q_0 = 32, p_0 = 2, N = 4, d_0 = k_0 = 2, t = 1$, outside (38). With the fixed negative-control inputs, it yields $v = (5, 31)$, $w = (11, 1)$, target bits $(0, 0)$ and decoded bits $(1, 0)$. The true lift fibers are respected. This confirms that the tests can detect a violation rather than presuming agreement.

The independent verifier does not import `polynomial_moddiv_a3.py`. It recomputes the 8,192 selected secret pairs through full convolutions and the normalized printed counter loop, reconstructs eight saved reference encryption traces, and checks 1,000 further signed window identities. It verifies 308 compatible field preimages in 250 sampled public experiments, 24 additional inverse fibers and 1,240 reconciliation cases using explicit sets. Recomputing the same fixtures is a consistency check, not an additional security benchmark.

Run the deterministic verification recipe from the extracted directory with `bash build_and_verify.sh`. The script invokes `verify_a3.py` and `verify_a3_independent.py`, then compares the regenerated JSON objects with the distributed results. The exact package commands are summarized again in Section 18. Neither verifier runs a network request. The stored synthetic reference traces contain test secrets solely to permit independent validation. The passive field solver receives only its public system. No LLL/BKZ campaign, large-dimensional rounded-secret recovery, CCA evaluation or quantum attack estimate is performed. The code checks functionality and the stated finite identities, not the computational assumption of Section 13.

17 Conclusion

Polynomial ModDiv contains three mathematically different layers that must not be conflated. P-WINDOW has a clean exact-window identity, but that same exactness lets any compatible public linear-system solution predict the common value over a finite field. P-ROUND retains the non-cyclic window structure while applying coefficientwise modular quotient extraction; its true deterministic remainders yield the residual identity (13), but a small circular residual does not by itself imply equality after a final quotient. The trailing-bit counter printed in the public proposal must first be applied to the integer-normalized value $\lfloor R_i/n \rfloor$; under the loop and prose readings analyzed above it still exhibits disagreement, representative dependence, undefined normalized-zero cases, and an erasure regime.

P-REF is different again. It uses nearest rounding, randomized inverse fibers, a long conditioned secret, a short public factor, reconciliation hints, and a complete encryption mechanism. Under

the stated computational MP-CLWR source assumption and the classical random-oracle model, the reference specialization has the conditional IND-CPA theorem proved above. That theorem is attributed to the published middle-product LWR line and does not establish security for historical P-ROUND.

The principal methodological conclusion is therefore negative but useful: commutativity, Toeplitz/Hankel structure, middle products, and the word "rounding" are algebraic similarities, not a security reduction. A valid transfer would have to preserve the complete public distribution, secret law, rounding rule, dimensions, and security task. No such bridge for historical polynomial ModDiv is claimed here.

18 Reproducibility

The accompanying package contains integer-exact Python reference code, an independent verifier that does not import the main arithmetic module, exact JSON result files, selected synthetic reference transcripts, and SHA-256 hashes. The deterministic checks cover signed window identities, finite-field passive prediction, rounded residual identities, fixed-divisor boundary witnesses, the normalized public trailing-bit counter readings, exact inverse-rounding fibers, reconciliation, weak middle-product associativity, universal-hash cases, reference encryptions, and the displayed parameter inequalities. These finite checks validate the stated instances and implementation paths; they do not prove the MP-CLWR assumption or any concrete security level.

Run from the extracted package directory:

```
bash build_and_verify.sh
bash compile_pdf.sh
```

The first command regenerates and compares both deterministic JSON result files. The second rebuilds the article from the distributed LaTeX source.

References

[1] S. Bouftass, "Multidimentional ModDiv public key exchange protocol," IACR Cryptology ePrint Archive, Paper 2021/701, 20 July 2021. https://eprint.iacr.org/2021/701

[2] S. Bai, K. Boudgoust, D. Das, A. Roux-Langlois, W. Wen, and Z. Zhang, "Middle-Product Learning with Rounding Problem and its Applications," ASIACRYPT 2019, LNCS 11921, pp. 55–81. IACR ePrint 2019/1001. https://eprint.iacr.org/2019/1001

[3] K. Boudgoust, *Theoretical Hardness of Algebraically Structured Learning With Errors*, PhD thesis, Université de Rennes 1, 2021. https://katinkabou.github.io/Documents/Thesis_Boudgoust_Final.pdf

Attribution boundary. Polynomial convolution, Toeplitz/Hankel representations, Gaussian elimination over finite fields, collision-based extraction, and random-oracle query arguments are established techniques. P-REF is attributed to the published middle-product LWR work above. The contributions of this article are the explicit separation of the three objects, the self-contained audit of the public polynomial ModDiv rules, the exact residual and boundary analysis, and the accompanying reproducible checks. No priority claim beyond the cited public record is made.